Huawei Access Node BAS-ACC-53 QoS Congestion Event Resolution SOP

Node: BAS-ACC-53 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_5037

Description

The Huawei Access Node BAS-ACC-53 is experiencing a QoS congestion event, manifesting as microbursts, interface tail drops, and QoS queue starvation. This failure domain is specific to this role and vendor equipment due to the Huawei Access Node's implementation of the IEEE 802.1p priority queuing mechanism, which is prone to congestion events when not properly configured. The underlying protocol mechanics of the Huawei Access Node's QoS architecture are based on the IEEE 802.1p standard, which assigns priority levels to frames based on their class of service (CoS). However, if the QoS policy is not correctly configured, it can lead to congestion events, causing microbursts, interface tail drops, and QoS queue starvation. Additionally, the Huawei Access Node's hardware mechanics, such as the TCAM (Ternary Content Addressable Memory) architecture, can also contribute to congestion events if not properly optimized.

Pre-requisite Checks

- Verify the site location and node ID to ensure that the issue is indeed on BAS-ACC-53.

- Check the redundant link status to ensure that there are no downstream network issues.

- Dispatch remote hands to the Basavanagudi POP to perform physical checks and ensure that the node is reachable.

Potential Root Causes

Cause	Probability
Misconfigured QoS policy	High
TCAM architecture not optimized	Medium
Hardware or software issues on the Huawei Access Node	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policy, Expected Output: QoS policy configuration, including class of service (CoS) and priority levels., Description: This command displays the QoS policy configuration, allowing the technician to verify that the policy is correctly configured and not contributing to the congestion event.

Command: display interface status, Expected Output: Interface status, including error counters and congestion statistics., Description: This command displays the interface status, allowing the technician to verify that there are no error counters or congestion statistics indicating a hardware or software issue on the Huawei Access Node.
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Command: display tcam statistics, Expected Output: TCAM architecture statistics, including hit rates and miss rates., Description: This command displays the TCAM architecture statistics, allowing the technician to verify that the hardware mechanics are not contributing to the congestion event.
```

Resolution Steps

Step 1: Step: Adjust QoS policy configuration, Command: qos policy {class-of-service} priority {priority-level}, Description: This step involves adjusting the QoS policy configuration to ensure that the policy is correctly configured and not contributing to the congestion event. The technician should verify that the policy is correctly configured using the diagnostic command "display qos policy".

Step 2: Step: Optimize TCAM architecture, Command: tcam optimize {hit-rate} {miss-rate}, Description: This step involves optimizing the TCAM architecture to ensure that the hardware mechanics are not contributing to the congestion event. The technician should verify that the TCAM architecture is correctly optimized using the diagnostic command "display tcam statistics".

Step 3: Step: Reset Huawei Access Node, Command: reset system, Description: This step involves resetting the Huawei Access Node to ensure that the node is functioning correctly. The technician should verify that the node is functioning correctly using the diagnostic command "display interface status".

Post-Check Verification

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Command: display qos policy, Expected Output: QoS policy configuration, including class of service (CoS) and priority levels., Description: This command verifies that the QoS policy configuration is correctly adjusted and not contributing to the congestion event.

Command: display interface status, Expected Output: Interface status, including error counters and congestion statistics., Description: This command verifies that there are no error counters or congestion statistics indicating a hardware or software issue on the Huawei Access Node.

Command: display tcam statistics, Expected Output: TCAM architecture statistics, including hit rates and miss rates., Description: This command verifies that the TCAM architecture is correctly optimized and not contributing to the congestion event.
```

Roll-back Procedures

- Step: Revert QoS policy configuration, Command: undo qos policy {class-of-service} priority {priority-level}, Description: This step involves reverting the QoS policy configuration to its previous state, in case the adjustment made in Step 1 was not successful.

- Step: Revert TCAM architecture optimization, Command: undo tcam optimize {hit-rate} {miss-rate}, Description: This step involves reverting the TCAM architecture optimization to its previous state, in case the optimization made in Step 2 was not successful.

- Step: Revert Huawei Access Node reset, Command: undo reset system, Description: This step involves reverting the Huawei Access Node reset to its previous state, in case the reset made in Step 3 was not successful.